

File Edit Format Options Window Execute ?



exp 4 emf.sci (C:\Users\pachi\Documents\exp 4 emf.sci) - SciNotes

exp 4 emf.sci

```
14 disp(r1);
15
16 // Surface charge density
17 rs = q/(4*pi*r^2);
18 disp("The surface charge density is (C/m^2):");
19 disp(rs);
20
21 // Volume charge density
22 rv = q/((4/3)*pi*r^3);
23 disp("The volume charge density is (C/m^3):");
24 disp(rv);
25
26 // Graph
27 x = linspace(0,10,50);
28
29 y = r1*exp(-0.1*x);
30 z = rs*exp(-0.1*x);
31 v = rv*exp(-0.1*x);
32
33 figure();
34
35 subplot(2,2,1);
36 plot2d3(x,y);
37 title("Line Charge Density");
38 xlabel("Length");
39 ylabel("rho_l");
40
41 subplot(2,2,2);
42 plot2d3(x,z);
43 title("Surface Charge Density");
44 xlabel("Length");
45 ylabel("rho_s");
46
47 subplot(2,2,3);
48 plot2d3(x,v);
49 title("Volume Charge Density");
50 xlabel("Length");
51
```

Scilab 2025.1.0 Console

File Edit Control Applications ?



Scilab 2025.1.0 Console

```
"The line charge density is (C/m):"
0.0001
"The surface charge density is (C/m^2)"
0.0265258
"The volume charge density is (C/m^3)"
26.525824
```

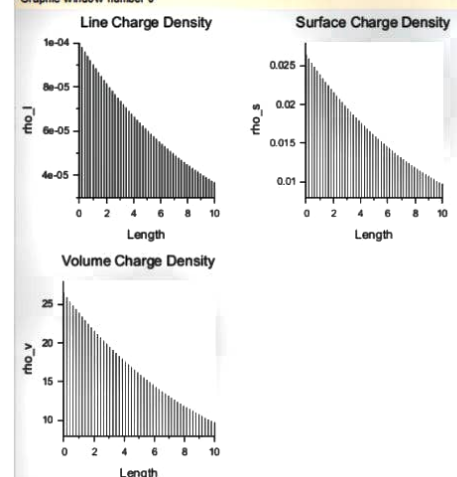
--> |

Graphic window number 0

File Tools Edit ?



Graphic window number 0



Exp 41 -

charge :- 3×10^{-6}

length :- 3 cm

radius :- 3 mm

i) Line charge

$$\rho_l = \frac{q}{L}$$

$$\rho_l = \frac{3 \times 10^{-6}}{0.03}$$

$$\rho_l = 1 \times 10^{-4} \text{ C/m}$$

ii) Surface charge

$$\rho_s = \frac{q}{4\pi r^2}$$

$$\rho_s = \frac{3 \times 10^{-6}}{4 \times 3.1416 \times (0.003)^2}$$

$$\rho_s = 0.0265 \text{ C/m}^2$$

iii) Volume charge density

$$\rho_v = \frac{q}{\frac{4}{3}\pi r^3}$$

$$\rho_v = \frac{3 \times 10^{-6}}{\frac{4}{3} \times 3.1416 \times (0.003)^3}$$

$$\rho_v = 26.53 \text{ C/m}^3$$